

[PLEXOS Cloud](#) > [PLEXOS Cloud](#) > [Command Line Interface \(CLI\)](#)

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Parquet V2 Overview

1. Overview

Parquet Version 2, or Parquet v2, is the newer Parquet schema used for PLEXOS Cloud solution files. It is the default schema for simulation runs.

If needed, you can still use the legacy Parquet v1 schema from the **Run Configuration** panel.

 **Warning:** Parquet v1 was deprecated in November 2024. Use Parquet v2 for new runs and automation workflows.

Benefits of Parquet v2

- **Enhanced compression** — reduces file size and improves read and write performance by using delta encoding for sorted timestamp columns.
- **Block reporting support** — adds a phase table that links reporting blocks to interval periods.
- **Compatibility with PLEXOS 10** — lets you open Parquet output like a native PLEXOS zip solution in PLEXOS Desktop.
- **Expanded data coverage** — includes additional data from the PLEXOS solution zip file, reducing the need to join output data with input data.

Request Parquet v2 in simulation enqueue

The simulation enqueue request supports a `SolutionOptions` object inside `SimulationOptions`. Use the `ParquetSchemaVersion` property to request schema version `1` or `2`.

Note: If `ParquetSchemaVersion` is not specified, existing scripts continue to work without changes.

Example request snippet:

```
{
  "SimulationOptions": {
    "Locale": 1033,
    "EnableRealTimeLog": true,
    "SolutionOptions": {
      "ParquetSchemaVersion": 2
    }
  },
  "SimulationData": [
    {
      "Type": "ChangesetDatabase"
    },
    {
      "Type": "TimeseriesZipped"
    }
  ],
  "SimulationEngine": {
    "EngineId": "c99f5481-61da-4199-8cec-e354ba17b7fe"
  }
}
```

Note: This is only a partial sample. It does not include every field required for a full enqueue request.

2. Toolkit changes for Parquet v2

Parquet v2 support also appears in other parts of the PLEXOS Cloud Toolkit.

- Commands under `plexos-cloud solution convert` include an option to specify the output schema version after conversion.

```
C:\Users\>plexos-cloud solution convert zip-to-parquet --help
CloudCLI.Commands.Solution.Convert.ConvertRawZipToParquetCommand

Converts the PLEXOS zip output database to a parquet format

Usage: plexos-cloud solution convert zip-to-parquet [options]

Options:
  -f|--format          Output format(json | csv | tsv | yaml | table | none)
                        Allowed values are: JSON, JSONL, CSV, None, Table, TSV, YAML.
                        Default value is: JSON.
  -q|--quiet           Turns off verbose messages from app
  -aut|--automation    Turns on developer friendly structured response messages from app
  -d|--outputDirectory Directory for command output to be written to (required)
  --overwrite          Overwrite any file that already exists
  --zipPath            Path to the PLEXOS zip output file (required)
  -?|-h|--help         Show help information.
```

- The `plexos-cloud solution list` response includes a `TypeVersion` property. For most types, this value is `NULL`. For Parquet solution types, the value is `1` or `2`.

```
{
  "SolutionId": "2f4d431a-24d3-4952-a2ee-a1f4295b8ddb",
  "StudyId": "255d8e5e-593d-4a28-afe1-3555e4f7e812",
  "ExecutionId": "afd0dc73-8db2-447d-80fb-5b675516c188",
  "SimulationId": "4e232a19-025b-43fe-8876-c7603bd9d280",
  "Type": "Parquet",
  "Status": "Complete",
  "LastUpdatedDate": "2023-10-05T23:50:05.9911131Z",
  "CreatedDate": "2023-10-05T23:50:05.991113Z",
  "ModelName": "Base",
  "TypeVersion": 1
}
```

Note: These Parquet schema version changes also apply to direct API integrations for simulation enqueue and solution type listing.

Schema changes in Parquet v2

Compression changes

- Data Parquet file partitions use **ZSTD compression** and **Delta Binary Packed** encoding.
- Reference: [Parquet delta binary packed encoding](#) ↗

Additional columns in `full_key_info`

- **ParentObjectCategoryName** — name of the parent object category in a collection.
- **ChildObjectCategoryName** — name of the child object category in a collection.
- **ParentClassLangId** — persistent Lang ID for the parent class across PLEXOS versions.
- **ChildClassLangId** — persistent Lang ID for the child class across PLEXOS versions.
- **CollectionLangId** — persistent Lang ID for the collection relationship.
- **PropertyLangId** — persistent Lang ID for the property across PLEXOS versions.
- **SeriesId** — key that represents a specific value for a specific Period ID.



Additional column in `t_class`

Logout



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PLEXOS Desktop

New table: `membership_info`

- Contains a subset of columns from `full_key_info` related to membership information.
- Includes parent and child object columns, class columns, and collection columns.
- Supports future use cases involving network topology membership dependencies.

New table: `phase`

- Maps simulation intervals to reporting periods by phase using `Interval_id`, `Phase_Type_Id`, and `Period_Id`.
- Supports block reporting use cases when Parquet data is flattened at the interval level.

Additional tables included from the original PLEXOS output database

- `t_class_group`
- `t_collection`
- `t_property`
- `t_unit`

These tables follow the same schema design as the PLEXOS database. They help reduce the need to access the input database for supporting details such as collection names and property units.

3. Entity relationship diagrams

Use the ERD reference below to review how PLEXOS manages both input data and solution data in Parquet format.

 PLEXOS Parquet-v2- ERD
Doc.pdf
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The **Input Database Schema** shows how model inputs are defined and related. The **Solutions Database Schema** shows how simulation results are stored, including time-series data linked to periods, phases, and keys.

4. Limitations

PLEXOS Cloud schema uses both start and end dates. If the date-time convention is misaligned, Parquet output can shift by one interval.

To avoid this issue, use **Beginning of Period** for the end date convention when working with cloud-generated Parquet solution files.

Warning: End of Period is supported, but it is not recommended for PLEXOS Cloud runs that use Parquet solution files.

Under **Date Time Convention**, select **Beginning of Period**.

If you are not using cloud-generated Parquet solution files, you can continue to use **End of Period**.

The screenshot shows the PLEXOS Cloud interface with the 'Report' tab selected. The 'Date Time Convention' dropdown is highlighted with a red box and a red arrow pointing to it, showing 'Beginning of Period' selected. Other settings visible include 'Solution File Formats' (Compressed XML (.zip) selected), 'Period Types' (Day selected), 'Stochastics' (Report Statistics and Save Each Sample unchecked), 'Filters' (Filter Objects (Interval) and Filter Objects (Summary) checked), and 'Flat Files' (Format: Datetime, Locale: (default)).

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